Version 1.0 Outline

This document will show each version of the 3D printed object that will be used to make the blinds move up and down automatically.

Background

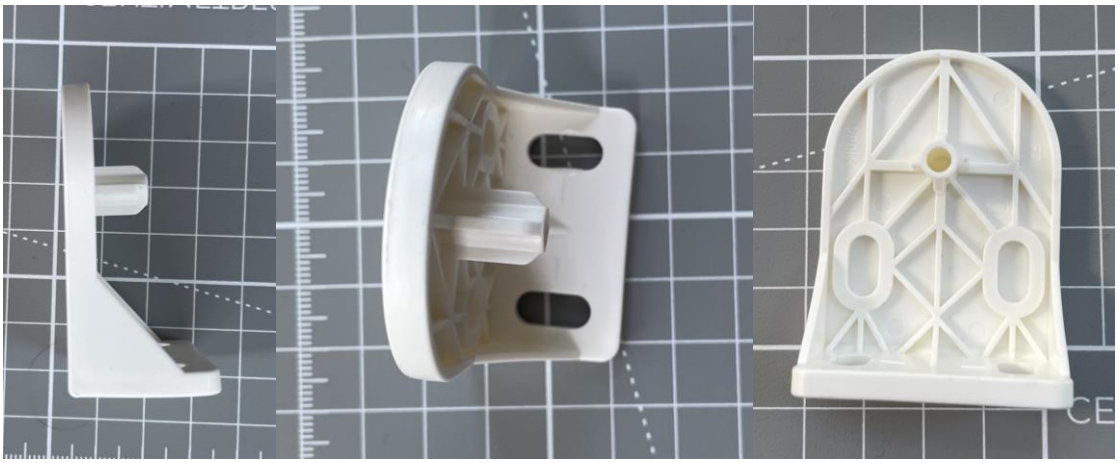
The approach to make automatic blinds is different from most as the blinds being used differ from most blinds in the way they pulled up and down. These blinds are from IKEA called FRIDANS Block-out roller blind, as per the reference image below there is no string.

Figure 1. FRIDANS blinds.

This project aims to add a motor to one of the support pieces to allow the motor to move the blinds up and down from the top mechanism and not the handle.

Original

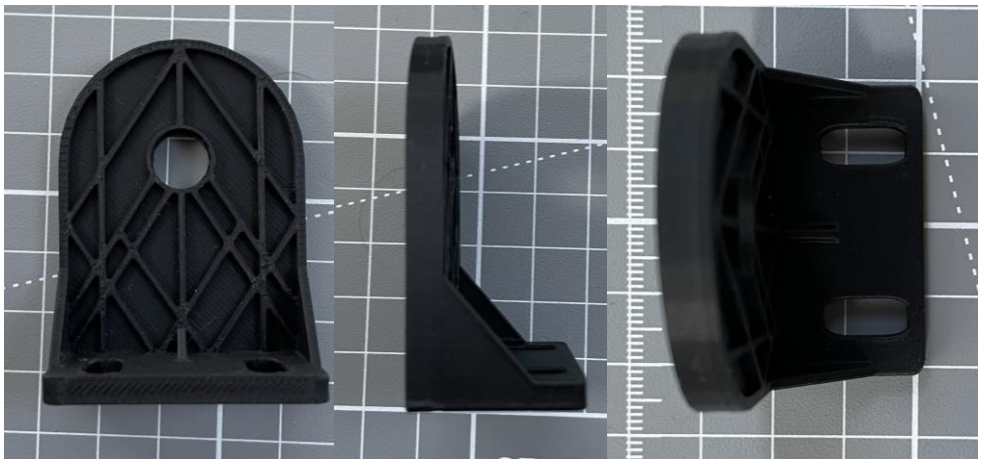
This is the support piece that will be modified. The cross-shaped drive that extrudes goes into a matching clearance at the top of the blinds, normally this stays still and allows the other side to rotate and allow the blinds to go up and down.



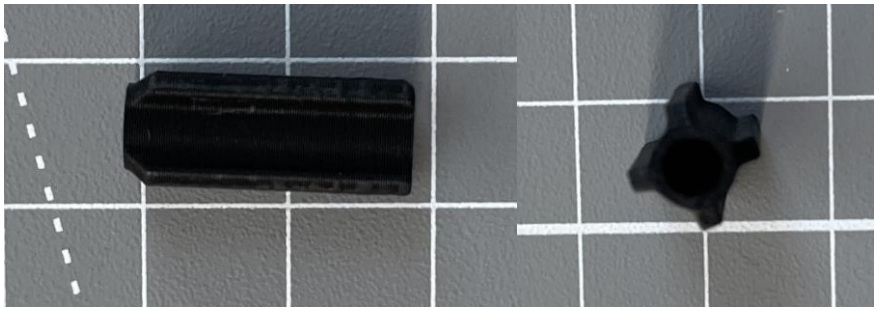
To allow the blind to be controlled by a motor the cross “screw” will be replaced by a stepper motor that will turn which will allow the motor to control the blinds.

Version 0.1 & 0.2

There are two different 3D prints for this however they are grouped together as the only difference between them is some clearance issues. There is also the cross-pattern drive that was printed.



The issue between v0.1 and v0.2 is that the hole the drive goes through does not have enough clearance. v0.2 is the same with just more clearance for the drive.

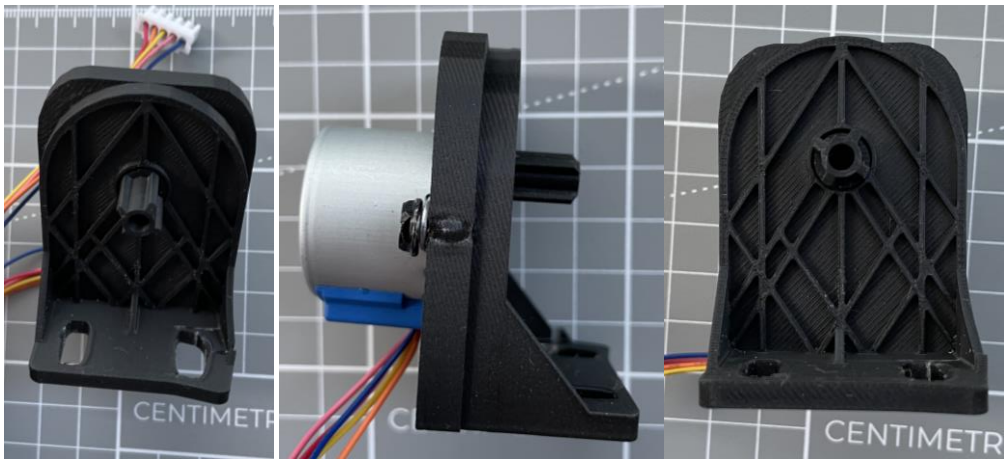


The drive will be connected to the motor and then fit through the support into the blind.

Version 1.0

This is the first version tested on the blinds. It was made a bit thicker to allow for through holes to be installed and therefore the motor to be connected to it.

An issue that arose with this is that it was not strong enough to withstand force of being screwed into the wall. It could be made thicker but switching from PLA to PTEG might be a good decision as well.



The stepper motor (28BYJ-48) was also not strong enough to fully open and close the blinds; however, this was known from the start and was only used as a stronger motor has not been ordered yet.